

12-WEEK ROADMAP

AI/LLM Engineering

RegIntel — Regulatory Intelligence Platform

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Track: AI/LLM Engineering

Programme: GetPostLabs Certification-Based Professional Internship

April 2026

Executive Summary

This document presents a 12-week execution plan for the AI/LLM Engineering workstream of the RegIntel (Regulatory Intelligence) platform at GetPostLabs. The plan is structured in three phases: Foundation & Domain Immersion (Weeks 1–4), Core AI/LLM Engineering (Weeks 5–8), and Advanced Capabilities & Production Readiness (Weeks 9–12).

The roadmap prioritises deep domain understanding before engineering execution, reflecting the philosophy emphasised across the RegIntel video series — that problem ownership and contextual knowledge are prerequisites for building meaningful AI systems. Each week has specific tasks, clear deliverables, and alignment with the certification track (AWS AI Practitioner, AWS ML Engineer, NVIDIA NCA-GENL, NVIDIA NCP-AAI).

My Approach & Philosophy

After watching all RegIntel videos carefully, three principles stood out that shape this entire plan:

- Domain first, technology second — Understanding what a SEBI circular means, how it relates to a Companies Act provision, and why a CA needs clause-level precision is more important than choosing between ChromaDB and Pinecone.
- The “big book” concept — RegIntel’s core insight is that hundreds of scattered documents become useful only when they are unified, cross-linked, and queryable as one interconnected knowledge base. The AI layer must preserve and leverage these connections.
- Multi-persona intelligence — The same regulatory fact needs to be communicated differently to a Chartered Accountant, a startup founder, and a home loan applicant. This is not just a UI concern; it fundamentally shapes how the LLM generates responses.

Phase Overview

Phase	Focus	Key Outcome
Phase 1 Weeks 1–4	Foundation & Domain Immersion	Deep domain understanding + working RAG prototype with regulatory documents
Phase 2 Weeks 5–8	Core AI/LLM Engineering	Multi-document reasoning, persona-aware responses, citation engine, fine-tuning benchmarks
Phase 3 Weeks 9–12	Advanced Capabilities & Production Readiness	AI agents, evaluation framework, platform integration, documentation & handover

Detailed Week-by-Week Plan

PHASE 1: FOUNDATION & DOMAIN IMMERSION (Weeks 1–4)

Week 1: Project Onboarding & Domain Deep-Dive

Tasks:

- Re-watch all 9 RegIntel videos; take structured notes on architecture, data flow, and team roles
- Map the Indian financial regulatory ecosystem: RBI, SEBI, IRDAI, PFRDA, NHB, IBBI — understand jurisdictions
- Study the RegIntel Part 5 breakdown of team roles; identify AI/LLM Engineer's touchpoints with Data Science, Platform, and Full-Stack teams
- Read the IMF references cited in videos to understand macroeconomic context
- Set up personal knowledge base (Notion/Obsidian) to document learnings, decisions, and open questions

Deliverable: Domain knowledge document covering regulatory bodies, document types, and how they interconnect. List of 10+ clarifying questions for the GetPostLabs team.

Week 2: Regulatory Document Taxonomy & Data Landscape

Tasks:

- Catalogue document types: circulars, master directions, notifications, gazettes, acts, rules, FAQs, press releases
- Identify source websites (RBI, SEBI, MCA, etc.) and analyse document formats (PDF, HTML, scanned images)
- Study how documents cross-reference each other (e.g., a SEBI circular referencing the Companies Act)
- Understand versioning challenges: amendments, superseded circulars, effective dates
- Explore existing open datasets or APIs for Indian regulatory data

Deliverable: Regulatory document taxonomy matrix (document type × regulator × format × frequency × cross-references).

Week 3: LLM Fundamentals & Tool Setup

Tasks:

- Deep-dive into transformer architecture, attention mechanisms, tokenization (BPE, SentencePiece)
- Study embedding models: OpenAI ada-002, Cohere embed, open-source alternatives (e5-large, BGE)
- Set up development environment: Python, LangChain/LlamaIndex, vector DB (Pinecone/Weaviate/ChromaDB)
- Run baseline experiments: embed sample regulatory text, perform similarity search
- Study prompt engineering techniques: few-shot, chain-of-thought, role-based prompting

Deliverable: Working local RAG prototype with 10–20 sample regulatory documents. Benchmark results for embedding quality.

Week 4: RAG Architecture Deep-Dive

Tasks:

- Study Retrieval-Augmented Generation end-to-end: query → retrieval → reranking → generation → citation
- Experiment with chunking strategies for regulatory text: fixed-size, semantic, section-based, hierarchical
- Implement metadata-enriched indexing: attach regulator, date, document type, section numbers to chunks
- Test retrieval quality with regulatory queries (e.g., “What are KYC norms for NBFCs?”)
- Study hallucination detection and grounding techniques for legal/regulatory content

Deliverable: Comparative analysis of 3+ chunking strategies with precision/recall metrics on regulatory queries. Architecture decision document for RegIntel’s RAG pipeline.

PHASE 2: CORE AI/LLM ENGINEERING (Weeks 5–8)

Week 5: Multi-Document Reasoning & Cross-Reference Resolution

Tasks:

- Build a cross-reference graph: link regulations that cite each other (Act → Rule → Circular → FAQ)
- Implement multi-hop retrieval: queries requiring information from 5–10 documents simultaneously
- Design prompt templates that instruct the LLM to synthesise across multiple sources with proper attribution
- Handle contradictions: when a newer circular supersedes an older one, ensure the LLM prioritises correctly
- Build a regulatory timeline module to track amendment history

Deliverable: Multi-document query engine capable of answering complex queries like “What changed in mutual fund regulations between 2020 and 2024?” with cited sources.

Week 6: Persona-Based Response Generation

Tasks:

- Design persona profiles as defined in RegIntel videos: CA, investment banker, startup founder, common citizen
- Build adaptive response templates that adjust depth, jargon level, and context based on user persona
- Implement persona detection from query patterns (or explicit selection)
- Create evaluation rubrics: does a CA get clause-level precision? Does a common user get plain-language explanations?
- Test with 20+ queries per persona; iterate on prompt engineering

Deliverable: Persona-aware query system with side-by-side comparison of responses for the same query across 4 personas.

Week 7: Citation Engine & Traceability

Tasks:

- Build a citation pipeline: every claim in a response maps back to a specific document, section, and clause
- Implement source verification: cross-check generated answers against retrieved chunks
- Design citation formats suitable for regulatory context (e.g., “RBI Master Direction on KYC, Para 3.2.1”)
- Build confidence scoring: flag low-confidence answers for human review
- Create an audit trail system for compliance and accountability

Deliverable: Citation-verified response system with confidence scores. Sample output showing query → answer → citations → confidence.

Week 8: Fine-Tuning & Domain Adaptation

Tasks:

- Prepare a fine-tuning dataset from regulatory Q&A pairs (manually curated + synthetically generated)
- Experiment with parameter-efficient fine-tuning: LoRA, QLoRA on open-source models (Llama 3, Mistral)
- Compare fine-tuned vs. RAG-only vs. hybrid approaches on regulatory queries
- Study instruction-tuning techniques specific to legal/regulatory language
- Document cost-performance trade-offs for different model sizes and approaches

Deliverable: Benchmarking report: fine-tuned model vs. RAG-only vs. hybrid on 50+ regulatory test queries. Recommendation on optimal approach for RegIntel.

PHASE 3: ADVANCED CAPABILITIES & PRODUCTION READINESS (Weeks 9–12)

Week 9: AI Agents for Regulatory Workflows

Tasks:

- Design agentic workflows: regulatory compliance checker, document comparison agent, impact analysis agent
- Implement tool-use patterns: agent calls retrieval tool, cross-reference tool, timeline tool as needed
- Build a “regulatory impact analyser”: given a new circular, identify all affected entities and existing rules
- Study ReAct, Plan-and-Execute, and other agent frameworks
- Implement guardrails: prevent the agent from making definitive legal interpretations

Deliverable: Working regulatory agent that can perform multi-step analysis (e.g., “If I start an NBFC, what are all compliance requirements?”).

Week 10: Evaluation Framework & Quality Assurance

Tasks:

- Build a comprehensive evaluation suite: factual accuracy, citation correctness, completeness, hallucination rate
- Create a golden test set: 100+ regulatory questions with verified ground-truth answers
- Implement automated evaluation using LLM-as-judge techniques
- Design human evaluation workflows for domain expert review
- Set up regression testing: ensure new changes don't degrade existing query quality

Deliverable: Evaluation dashboard with metrics across all dimensions. Documented test suite that can be run on every model/pipeline update.

Week 11: Platform Integration & API Design

Tasks:

- Design the AI/LLM service API: endpoints for query, persona selection, citation retrieval, feedback
- Coordinate with Platform Engineering on deployment architecture (containerisation, scaling, monitoring)
- Coordinate with Full-Stack team on frontend integration: streaming responses, citation rendering, feedback UI
- Implement rate limiting, caching, and cost optimisation for LLM API calls
- Design the feedback loop: user corrections flow back into improving retrieval and generation

Deliverable: API specification document. Integration test suite. Cost projection for production traffic.

Week 12: Documentation, Handover & Future Roadmap

Tasks:

- Write comprehensive technical documentation: architecture decisions, pipeline details, deployment guide
- Create a "RegIntel AI/LLM Playbook" for future team members
- Document all experiments, what worked, what didn't, and why
- Propose a 6-month forward roadmap: multilingual support, real-time regulatory tracking, advanced analytics
- Prepare a demo and presentation for the GetPostLabs team

Deliverable: Complete documentation package. 6-month forward roadmap. Demo-ready system showcasing end-to-end capabilities.

Certification Alignment

The roadmap is designed to align practical project work with the four certifications offered in the AI/LLM Engineering track. Each certification’s concepts are reinforced through hands-on RegIntel work during the corresponding weeks.

Certification	Weeks	RegIntel Alignment
AWS AI Practitioner	1–4	Foundational AI/ML concepts, responsible AI, choosing the right approach
AWS ML Engineer	5–8	ML pipelines, model training, deployment, monitoring in production
NVIDIA NCA-GENL (Generative AI)	3–10	LLM architecture, RAG, prompt engineering, fine-tuning, evaluation
NVIDIA NCP-AAI (AI Agents)	9–12	Agentic AI, tool use, multi-step reasoning, guardrails

Cross-Team Collaboration Model

As highlighted in RegIntel Part 5, the AI/LLM Engineer does not work in isolation. Here is how I plan to collaborate with each team throughout the 12 weeks:

Team	Collaboration Points	Timing
Data Science	Document parsing quality, entity extraction, taxonomy validation, evaluation metrics design	Continuous, heavy in Weeks 1–4
Platform Engineering	Deployment architecture, vector DB infrastructure, API gateway, monitoring, scaling	Weeks 9–12 primarily
Full-Stack Engineering	Frontend integration, streaming response rendering, citation UI, feedback collection interface	Weeks 10–12

Key Problems Identified & My Proposed Approaches

Based on my analysis of the RegIntel videos and the regulatory landscape, these are the core AI/LLM engineering challenges I anticipate:

1. Fragmented, inconsistent document formats

Challenge: Regulatory documents come in PDFs (some scanned), HTML pages, and structured formats. Quality of text extraction directly impacts LLM performance.

Approach: Work with Data Science team on multi-format parsing pipeline. Implement quality checks on extracted text. Use OCR + layout-aware extraction for scanned documents.

2. Cross-document reasoning at scale

Challenge: A single user query might require synthesising information from 5–10 different regulations across multiple regulators. Standard RAG retrieves chunks, not relationships.

Approach: Build a regulatory knowledge graph alongside vector search. Use graph-augmented RAG where the retriever traverses document relationships, not just semantic similarity.

3. Hallucination in high-stakes domain

Challenge: Regulatory advice has real consequences. A hallucinated clause number or incorrect interpretation could lead to compliance failures.

Approach: Implement mandatory citation verification: every generated claim must map to a retrieved source. Confidence scoring flags uncertain answers. Never let the system present itself as legal advice.

4. Temporal validity of regulations

Challenge: Regulations are amended, superseded, and versioned. The system must know which version is currently in force.

Approach: Implement a temporal metadata layer: each document chunk carries effective dates, amendment history, and supersession links. The retriever filters by current validity before passing to the LLM.

5. Persona-appropriate responses without losing accuracy

Challenge: Simplifying for a common user must not sacrifice factual correctness. A startup founder needs actionable steps, not just legal citations.

Approach: Two-stage generation: first generate a factually complete internal response, then transform it through persona-specific prompts that adjust language and emphasis while preserving all facts.

Technology Stack (Proposed)

Category	Tools & Technologies
LLM Providers	OpenAI GPT-4o, Anthropic Claude, open-source (Llama 3, Mistral) for fine-tuning experiments
Orchestration	LangChain / LlamaIndex for RAG pipelines; LangGraph for agentic workflows
Vector Database	ChromaDB (development), Pinecone or Weaviate (production evaluation)
Embeddings	OpenAI text-embedding-3-large, BGE-large-en (open-source alternative)
Fine-Tuning	Hugging Face Transformers, PEFT (LoRA/QLoRA), Weights & Biases for experiment tracking
Evaluation	RAGAS, DeepEval, custom regulatory accuracy metrics
Infrastructure	Docker, AWS (aligning with certification track), GitHub Actions for CI/CD
Knowledge Graph	Neo4j or NetworkX for regulatory cross-reference graph

Success Metrics

By the end of 12 weeks, the AI/LLM Engineering workstream should demonstrate:

- Retrieval precision above 85% on regulatory queries across all document types
- Citation accuracy above 95%: every claim in a generated response traces back to a real document and clause
- Hallucination rate below 5% on the golden test set of 100+ regulatory questions
- Persona satisfaction: responses rated as “useful and appropriately detailed” by test users in each persona category
- Multi-document queries: ability to correctly synthesise answers requiring 5+ source documents
- All four certifications either completed or on track for completion

Closing Note

This roadmap reflects not just a technical plan but a commitment to understanding the regulatory domain deeply before engineering solutions. The RegIntel video series makes it clear that the strongest contributions come from team members who own the problem, not just the technology. I’m excited to bring this mindset to the project and to learn from the GetPostLabs team throughout the journey.

I welcome feedback on this plan and am ready to adapt it based on the team’s priorities and the project’s current state. This is a living document — the best version of it will be the one we build together.